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Lactobacillus curvatus has a glucose transport system homologous to the mannose family of phosphoenolpyruvate-dependent phosphotransferase systems

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In *Lactobacillus curvatus*, a phosphoenolpyruvate:mannose phosphotransferase system (mannose-PTS) has been characterized and it was shown to be involved in glucose and mannose transport, but no glucose-specific PTS activity could be detected. A 2·1 kb DNA fragment amplified by PCR from the *L. curvatus* genome was sequenced. Sequence analysis showed four ORFs which could encode proteins similar to PTS transporters EIIA, EIIB, EIIC and EIID of the mannose class. The expression of the *manB* gene (encoding EIIB) from *L. curvatus* in a mutant of *Lactobacillus sake* impaired in EII^{Man} activity restored this activity. Furthermore, this DNA fragment complemented the regulatory function of LevE (EIIB) in a *Bacillus subtilis levE*-deficient mutant, suggesting that the protein encoded by *manB* could also play a regulatory role in *L. curvatus*.

Keywords: *Lactobacillus curvatus*, mannose phosphotransferase system, complementation of *Bacillus subtilis* LevE activity

INTRODUCTION

Species belonging to the genus *Lactobacillus* are found in a wide variety of fermented food products derived from milk, meat, cereals and vegetables. In particular, the species *Lactobacillus sake* and *Lactobacillus curvatus* are essential agents in dry-cured meat products (Hammes *et al.*, 1990). In the meat mixture, added carbohydrates and certain amounts of amino acids are available to the bacterial cells. The metabolism of these compounds can generate a large amount of organic acids and flavour components. However, glucose, the most rapidly metabolized carbon source, can affect the production of enzymes, extracellular polysaccharides and secondary metabolites. This regulatory role is, to a great extent, dependent on the glucose-specific transport mechanisms present in these micro-organisms (Postma *et al.*, 1993).

Glucose can enter the bacterial cell via two phosphoenolpyruvate (PEP)-dependent sugar phosphotransferase systems (PTS) specific for glucose or mannose (EII^{Glc} or EII^{Man}, respectively). The PTS comprises two cytosolic proteins, Enzyme I and HPr, which are common to all sugar transport systems, and a family of sugar-specific EII elements, consisting of three or four domains (A, B, C or D) that can be part of a single protein or independent polypeptides (Saier & Reizer, 1992). According to the alignments of the deduced amino acid sequences of known EII proteins, Postma *et al.* (1993) grouped them into four distinct families: glucose-PTS, mannose-PTS, lactose-PTS and mannitol-PTS. The mannose-PTS class includes the mannose-specific EII of *Escherichia coli* (EII^{Man}) (Erni *et al.*, 1987), the sorbose-specific EII (EII^{Sor}) of *Klebsiella pneumoniae* (Wehmeier & Lengeler, 1994), the fructose-specific EII (EII^{Lev}) of *Bacillus subtilis* (Martin *et al.*, 1990) and the putative *N*-acetylgalactosamine-specific EII (EII^{Aga}) of *E. coli* (Reizer *et al.*, 1996). A particular case was found in *Corynebacterium glutamicum* in which the sugar specificity of the glucose transport system corresponded to that described for the EII^{Man} of *E. coli* (Lee *et al.*, 1993). However, the sequence alignment revealed that it belonged to the sucrose-β-glucoside subgroup of the glucose class (Lee *et al.*, 1994; Postma *et al.*, 1993). No other sequences of genes encoding EII proteins of the mannose class have been described. The mannose class EII complex is composed of two proteins (IIC and IID)

Abbreviations: 2-DG, 2-deoxy-D-glucose; αMG, methyl α-D-glucopyranoside; PEP, phosphoenolpyruvate; PTS, PEP:sugar phosphotransferase system.

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that form the transmembrane element and one or two associated cytoplasmic enzymes. In the EII^{Man} of *E. coli*, there is a single soluble protein with two domains (IIAB) (Erni *et al.*, 1987), whilst two separate polypeptides (IIA and IIB) are found in EII^{Lev} (Martin *et al.*, 1990) and EII^{Sor} (Wehmeier & Lengeler, 1994). Furthermore, mutants in different EII^{Lev} elements of *B. subtilis* have been isolated (Kunst *et al.*, 1977) in which the expression of the levanase operon was constitutive; in fact, elements IIA and IIB act as negative regulators of the levanase operon (Martin *et al.*, 1990).

In lactic acid bacteria, EII^{Man} has been proposed to be the main phosphotransferase-mediated transport system for glucose, for which regulatory functions have been described (Chassy & Thompson, 1983; Abe & Uchida, 1989; Veyrat *et al.*, 1994; Gauthier *et al.*, 1990). In *Lactococcus lactis* and *Pediococcus halophilus*, EII^{Man} has been shown to be involved in inducer exclusion/expulsion events as well as in catabolite repression of the genes necessary for the assimilation of lactose, galactose and xylose (Thompson & Chassy, 1985; Abe & Uchida, 1989). In *Lactobacillus casei*, elements of EII^{Man} may be involved in the control of lactose and ribose metabolism (Veyrat *et al.*, 1994), and in *Streptococcus salivarius* and *Streptococcus vestibularis* EII^{Man} may control sugar uptake (Gauthier *et al.*, 1990).

The aim of the work reported in this article was the biochemical and molecular characterization of glucose transport in *L. curvatus* and its involvement in the regulation of metabolic pathways leading to an efficient utilization of carbon sources. PEP-dependent PTS activity assays were carried out to test substrate specificity to determine the type of EII glucose transporter and its level of activity in bacterial cells which were grown on other sugars. The responsible genes were then amplified by PCR with the aid of degenerate primers obtained from the alignment of known sequences of members of the mannose EII family. The complete nucleotide sequence of the genes encoding EIIA, EIIB and EIIC (*manA*, *manB* and *manC*) and the partial primary structure of *manD*, encoding EIID of *L. curvatus*, are reported. The involvement of the PTS system in glucose and mannose transport could be demonstrated using a cloned fragment that contained the gene encoding $EIIB^{Man}$ from *L. curvatus* to complement an EII^{Man}-deficient mutant of the closely related species *L. sake*. Finally, complementation experiments with an $EIIB^{Lev}$-deficient mutant of *B. subtilis* suggest a possible regulatory function for the $EIIB^{Man}$ protein in *L. curvatus*.

METHODS

Bacterial strains, culture conditions and plasmids. *Lactobacillus curvatus* CECT 904, *Lactobacillus sake* 160*1K and RV37 (Lauret *et al.*, 1996) and *Lactobacillus casei* ATCC 393, cured of pLZ15, were used in this work. These strains were grown at 30 °C or 37 °C under static conditions on MRS (Oxoid), Fermentation MRS medium (ADSA-MICRO) or MCD medium (Lauret *et al.*, 1996) supplemented with 0·5 % sugar, as indicated. *Escherichia coli* DH5α [F^- *endA1 hsdR17 gyrA96 thi1 recA1 relA1 supE44* Δ*lacU169* (ϕ80 *lacZ* DM15)] and *E. coli* Novablue (Novagene) were used as hosts in cloning procedures. These cells were grown on Luria–Bertani (LB) medium (Miller, 1992) and ampicillin (100 μg ml^{-1}) was added for plasmid selection. *Bacillus subtilis* QB5072 (*trpC2 levE7 sacC'–lacZ erm*) was kindly provided by Professor G. Rapoport (Institut Pasteur, France). This strain was grown on LB or basal MMCH medium (Martin *et al.*, 1990) at 37 °C with vigorous aeration. Bacterial growth was carried out in the presence of 1 μg erythromycin ml^{-1} and 25 μg lincomycin ml^{-1}. For agar plates, 1·7 % (w/v) agar was added to the medium.

The plasmid used in *E. coli* experiments was pT7Blue-T-vector (Novagen). pGAL9 (Pérez-Martínez *et al.*, 1992) was used as a shuttle vector for cloning purposes in *B. subtilis*. This vector contains a secretory signal sequence (AL9) carrying transcriptional and translational initiation sequences in addition to the bacteriophage SPO2 promoter located upstream of the truncated α-amylase gene, the broad-host-range pWV01 replicon (Kok *et al.*, 1984) and the erythromycin-resistance gene from pE194. *B. subtilis* QB5072 transformants were selected on LB plates containing 200 μg erythromycin ml^{-1}.

DNA manipulation. The purification of genomic DNA from *Lactobacillus* strains was performed according to Posno *et al.* (1991) with some modifications. Cells were grown in 10 ml MRS to early stationary phase, harvested by centrifugation and washed with 10 ml 50 mM EDTA. The pellet was resuspended in 600 μl lysis buffer (10 %, w/v, PEG 20000 in 15 mM Tris/HCl, pH 8; 5 mM disodium maleic acid; 5 mM $MgCl_2$) supplemented with lysozyme, at a final concentration of 5 mg ml^{-1}, and mutanolysin at 20 U ml^{-1}. After 1 h at 37 °C, cells were collected by centrifugation, resuspended in 600 μl 20 mM Tris/HCl (pH 8) and lysed with 60 μl 10 % (w/v) SDS in 50 mM Tris/HCl, 20 mM EDTA (pH 8). The following steps were performed as standard protocols (Sambrook *et al.*, 1989).

DNA fragments were recovered from agarose gels using a Sephaglas BandPrep kit (Pharmacia). The DNA probe used in Southern blot experiments and RNA dot-blotting was labelled by the non-radioactive random primed method (DIG DNA Labelling and Detection kit; Boehringer). The temperature of Southern blot hybridization was 55 °C. The RNA dot-blotting hybridization was carried out at 42 °C in the presence of 50 % (v/v) formamide. The following washing and detection were carried out according to the manufacturer's instructions. Restriction and modifying enzymes were also used according to the recommendations of the manufacturers. General cloning procedures were performed according to Sambrook *et al.* (1989). Preparation of competent cells and transformation of *B. subtilis* and *L. sake* were carried out as described previously by Bron & Luxen (1985) and Berthier *et al.* (1996).

DNA was sequenced by the dideoxy chain-termination method (Sanger *et al.*, 1977) using a T7 Sequencing kit (Pharmacia). Synthetic DNA primers used for sequencing purposes were provided by Pharmacia. Analysis of DNA and protein sequence data was carried out with the following commercial packages: CLUSTAL (Higgins & Sharp, 1989) and GCG (Genetics Computer Group). EMBL, GenBank, PIR and SWISS-PROT were the databases used for DNA and protein similarity searches.

Amplification by PCR. In order to amplify genes encoding EII^{Man} from *L. curvatus* by PCR, various synthetic primers were designed: man01 (5'-CTCAGAAAAGATACAAGC), man02 (5'-TGGTTCCAGAGATAACATCC), man11 (5'-GGAGGSASSSCCATATAAYGC), man12 (5'-TTCAGCAACACCATCAGAAAC), man21 (5'-ATYGATGATCGBTTTATTCAYGG), man22 (5'-CAARCCMACCAASGTACANGC) and man42 (5'-GGATCNCCNACNCCNGC), where B could be C,

G or T; M could be A or C; N could any nucleotide; R could be A or G; S could be C or G and Y could be C or T. The amplification reaction contained 0·1 μg of *L. curvatus* genomic DNA as template, 100 pmol of each primer, 1 U *Taq* DNA polymerase and the standard 10× reaction buffer provided by the manufacturer (Boehringer). The reaction was carried out in 30 cycles of 94 °C for 30 s, 55 °C for 30 s and 70 °C for 1 min, followed by a final primer extension of 72 °C for 5 min. The reactions were performed in a Techne PHC-3 thermal cycler.

Total RNA purification and dot-blot analysis. Buffers for RNA extraction were prepared with diethyl pyrocarbonate (DEPC)-treated water as described by Sambrook *et al.* (1989). *L. curvatus* was grown to early stationary phase (16 h) in 100 ml MRS fermentation medium with 0·5% glucose, mannose, fructose, ribose, maltose or galactose as carbon sources. Cells were collected by centrifugation, washed with ice-cold DEPC-treated water and resuspended in 1/10 vol. of killing solution (0·4 M ammonium acetate; 0·01 M magnesium acetate; 25%, w/v, sucrose) supplemented with 2 mg lysozyme ml^{-1} and 10 U mutanolysin ml^{-1}. The cellular suspensions were incubated at 37 °C for 1 h and then harvested by centrifugation at 4 °C. The pellet was resuspended in 5 ml lysis buffer (0·2 M sodium acetate; 1%, w/v, SDS; 0·01 M EDTA; pH 8) containing proteinase K (100 μg ml^{-1}) and incubated for 30 min at 37 °C. Phenol/chloroform/isoamyl alcohol (10 ml) was then added. The nucleic acids, in the aqueous phase, were collected and precipitated by adding lithium chloride to a final concentration of 2 M and 2 vols ethanol and kept at −20 °C overnight. A pellet was obtained by centrifugation at 4 °C for 30 min, washed with 70% (v/v) ethanol and resuspended in DNase I buffer (20 mM Tris/HCl, pH 8; 10 mM $MgCl_2$) with 10 U RNase-free DNase (Boehringer). After 1 h at 37 °C, another extraction with phenol/chloroform/isoamyl alcohol was performed and RNA was precipitated with ethanol. The RNA was then resuspended in 0·1 ml DEPC-treated water. A_{260} and A_{280} measurements were taken to estimate the RNA concentration and purity (Hewlett Packard HP8452A spectrophotometer). Dot blotting was performed using nylon membrane (Hybond-N; Amersham) and a Bio-Dot Microfiltration Apparatus (Bio-Rad).

PEP-dependent PTS activity. PTS activity, estimated by consumption of PEP in the presence of mannose, 2-deoxy-D-glucose (2-DG), glucose, fructose, ribose and methyl α-D-glucopyranoside (α-MG), was quantified according to Chassy & Thompson (1983).

Enzyme assays. The procedure used to measure the amount of fructose 6-phosphate formed from PEP was a modification of that described by Abe & Uchida (Veyrat *et al.*, 1994).

The amount of fructose 1-phosphate formed in permeabilized cells in the presence of fructose and PEP was determined according to Sanz *et al.* (1994). The sugar phosphate was dephosphorylated using DOG^{R2} phosphatase, kindly provided by Dr Prieto (IATA, CSIC, Valencia, Spain) (Randez-Gil *et al.*, 1995), and the release of inorganic phosphate was measured as described by Dittmer & Wells (1969).

B. subtilis β-galactosidase activity was estimated on LB plates supplemented with X-Gal and 0·2% fructose. Quantitative studies were performed as described by Miller (1992). Cells were grown in liquid MMCH medium supplemented with 0·2% fructose or glycerol, washed twice with ice-cold water and resuspended in β-galactosidase buffer (60 mM $Na_2HPO_4.7H_2O$, 40 mM $NaH_2PO_4.H_2O$, 10 mM KCl, 1 mM $MgSO_4.7H_2O$, 50 mM β-mercaptoethanol) to give an OD_{600} of 1–2. Cell-free extracts were obtained by shaking 1 ml of bacterial suspension in a vortex with 1 g glass beads (0·1 mm diameter). Disruption was achieved after four periods of 30 s with 1 min interval on ice. The mixture was then centrifuged at 2600 *g* for 15 min. Crude extracts obtained were transferred to fresh tubes and maintained on ice until use. Protein concentration was determined according to Bradford (1976), using bovine serum albumin as standard. Specific β-galactosidase activity is expressed as Miller units (mg protein)$^{-1}$ (Miller, 1972).

RESULTS AND DISCUSSION

PEP-dependent PTS activity

PTS-mediated sugar phosphorylation activity in *L. curvatus* was quantified by measuring the consumption of PEP in the presence of a given sugar by permeabilized cells (Table 1). When cells were grown on glucose, mannose or fructose, the consumption of PEP in the presence of glucose or mannose was similar in all cases and they all showed 2-DG phosphorylation activity but no corresponding activity on α-MG. In the case of ribose-grown cells, glucose and mannose phosphorylation activity was higher than that found in cells grown on hexoses. This indicates that, although the system may be partially repressed by hexoses, its constitutive level of activity was high. Ribose-PTS activity was very low in all cases (5% of the activity found on glucose) and, since cells could normally grow on ribose, a non-PTS transport system must be present in *L. curvatus*. The pattern of sugar specificity of this transport system was similar to that found in the EII^{Man} of *E. coli* and also resembled the pattern found by Veyrat *et al.* (1994) in *L. casei*.

Fructose phosphorylation

Table 1 shows that fructose phosphorylation by glucose- or mannose-grown cells was very low, but that by fructose-grown cells was slightly higher. In order to confirm the inability of *L. curvatus* EII^{Man} to phosphorylate fructose, two phosphorylation products were quantified, fructose 6-phosphate and fructose 1-phosphate. Fructose 6-phosphate is normally formed by EII^{Man} of *E. coli* and EII^{Lev} of *B. subtilis* (Erni *et al.*, 1987; Martin *et al.*, 1990) but the C-1 of fructose is phosphorylated yielding fructose 1-phosphate in EII^{Sor} of *K. pneumoniae* (Wöhrl & Lengeler, 1990). No fructose 6-phosphate or fructose 1-phosphate was detected in cells preadapted with fructose, glucose or mannose. According to these data, *L. curvatus* EII^{Man} could be an unusual member of the group, as it has a low affinity for fructose.

Distinct fructose transport systems may be present in *L. curvatus*, since this micro-organism can efficiently grow on fructose and we could show that a higher EII^{Fru} activity was found in fructose-grown cells.

Cloning of genes encoding EII^{Man}, and sequence analysis

From the sequence alignment of known EII proteins (Postma *et al.*, 1993), different synthetic oligonucleotides (man11, man12, man21, man22 and man42) were designed from conserved regions of the mannose-EII family. Initially, three DNA fragments of 340, 600 and 860 bp

Table 1. PTS activity in permeabilized cells of *L. curvatus*

PTS activity is expressed as nmol PEP consumed min^{-1} (mg dry wt)$^{-1}$.

Growth on:	PTS activity in the presence of:					
	Glucose	Mannose	Fructose	Ribose	α-MG	2-DG
Glucose	142·9	161·9	12·7	15·9	12·7	66·7
Mannose	189·8	206·8	5·6	8·5	5·6	82·1
Fructose	149·5	155·8	31·2	9·3	9·3	68·5
Ribose	326·6	361·3	17·3	17·3	11·6	130·1

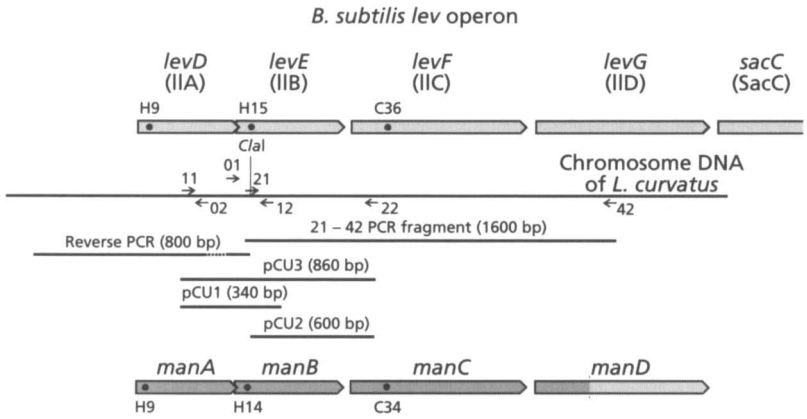


Fig. 1. Strategy of cloning of the *man* operon from *L. curvatus*. The synthetic oligonucleotides used for PCR amplification are indicated by arrows. pCU1, pCU2 and pCU3 plasmids were obtained by cloning the indicated amplified fragment in pT7Blue T-vector. IIA, IIB, IIC and IID elements correspond to the gene order of the levanase operon in *B. subtilis*. A putative structure of the EII^{Man} operon from *L. curvatus* is shown.

were amplified with these primers by PCR using *L. curvatus* CECT 904 total DNA as template (Fig. 1). They were cloned in pT7Blue T-vector to obtain pCU1, pCU2 and pCU3, respectively, and the three cloned fragments were sequenced. Then, in order to obtain the downstream region, another fragment of 1600 bp was amplified from man21 and man42. Finally, a DNA fragment of 800 bp including the 5′-end of *manA* was amplified by reverse PCR. The total DNA of *L. curvatus* was digested with the restriction enzyme *Cla*I, ligated with T4 ligase and used as template for PCR amplification with the specific primers man01 and man02. Sequencing was performed directly from the PCR product.

Fig. 2 shows the longest nucleotide sequence obtained from *L. curvatus*. Primary structure analysis revealed four ORFs (ORF1–4), although ORF4 was incomplete. The first two ORFs could encode proteins of 143 and 163 aa, respectively, and ORF3 could encode a protein of 271 aa. Detailed analysis showed Shine–Dalgarno sequences 7–10 nucleotides upstream of the putative start codons, ATG, in all four ORFs. The first codon of ORF2 overlapped with the stop codon (TGA) of ORF1, which could indicate translational coupling of ORF1 and ORF2. The intercistronic distance between ORF2 and ORF3 was 19 bp and that between ORF3 and ORF4 was 12 bp.

The deduced amino acid sequences were tested for similarities with sequences in the databases. Higher sequence similarities were found to EII belonging to the mannose class. Alignments with the deduced amino acid sequences of *E. coli* EII^{Man} (Erni *et al.*, 1987), *B. subtilis* EII^{Lev} (Martin *et al.*, 1990), *K. pneumoniae* EII^{Sor} (Wehmeier & Lengeler, 1994) and *E. coli* EII^{Aga} (Reizer *et al.*, 1996) are shown in Fig. 3. The calculated percentages of similarity and identity are shown in Table 2. The sequence of $EIIA^{Aga}$ from *E. coli* was not included in the comparison, as it is not included in the operon. In general terms, the degree of similarity between the EII^{Man} proteins of *L. curvatus* and the EII^{Lev} proteins of *B. subtilis* was higher than to any other EII complex. The trend of sequence divergence between the different EIIA proteins, also found between $EIIA^{Man}$ of *L. curvatus* and the other EIIA compared, as also shown by Reizer *et al.* (1996), was also noticeable. Therefore, it can be considered that the sequenced fragment of *L. curvatus* DNA encodes an operon with a structure similar to the levanase operon of *B. subtilis* and the sorbose operon of *K. pneumoniae* and, considering their structure, they could also have translational coupling (Martin *et al.*, 1990; Wehmeier & Lengeler, 1994). However, a difference between these two organisms is that the expression of these genes in *L. curvatus* seems to be constitutive (see below) as is the expression of the genes encoding EII^{Man} of *E. coli*. Sequence analysis also showed that proteins IIA^{Man} and IIB^{Man} of *L. curvatus* have a histidine residue (His-9 and His-14, respectively) that could be involved in the transfer of the phosphate group to the carbohydrate, as occurs in the other members of the family. There are also highly

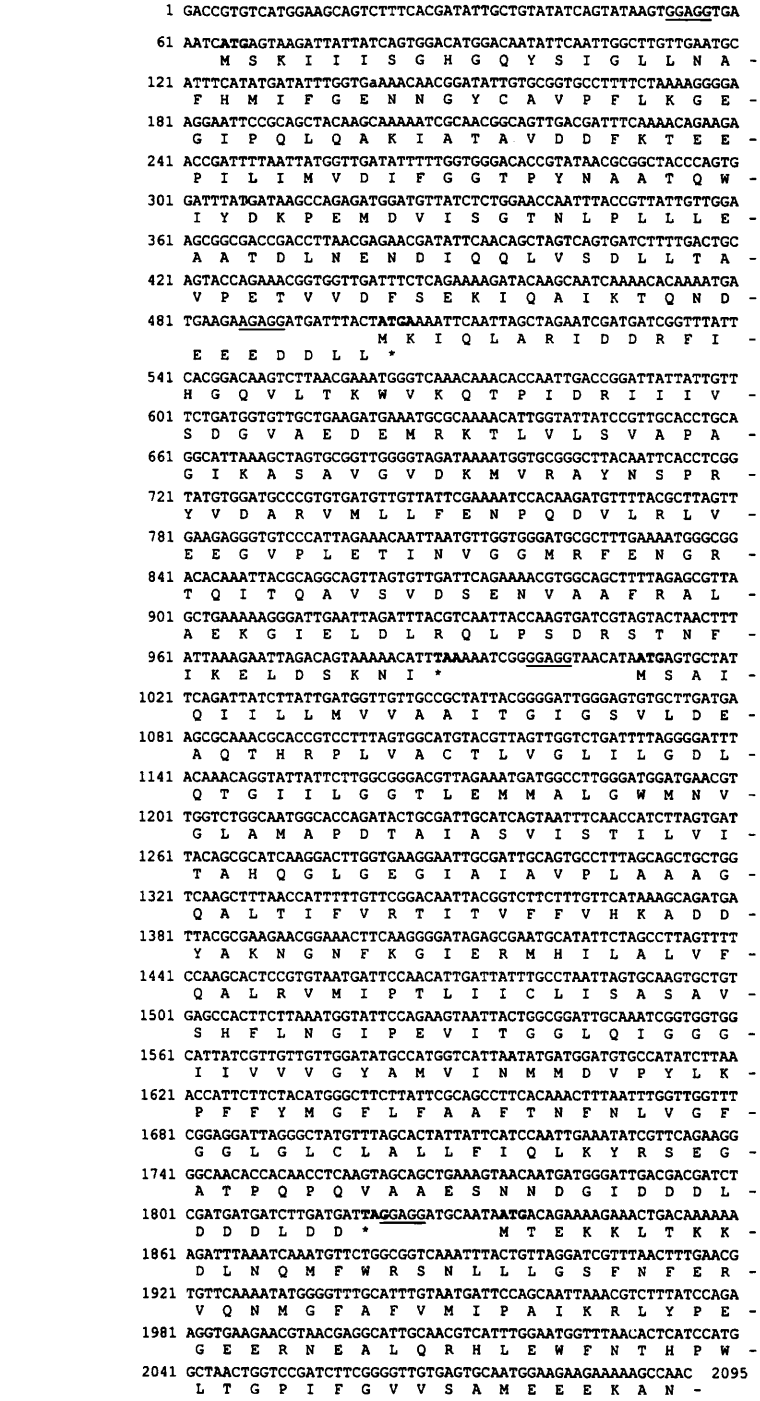


Fig. 2. Nucleotide sequence of the *L. curvatus* DNA fragment containing four ORFs. The deduced amino acid sequences are written below the corresponding nucleotide codons. Sequences resembling ribosome-binding sites are underlined. Putative start codons are indicated in bold type.

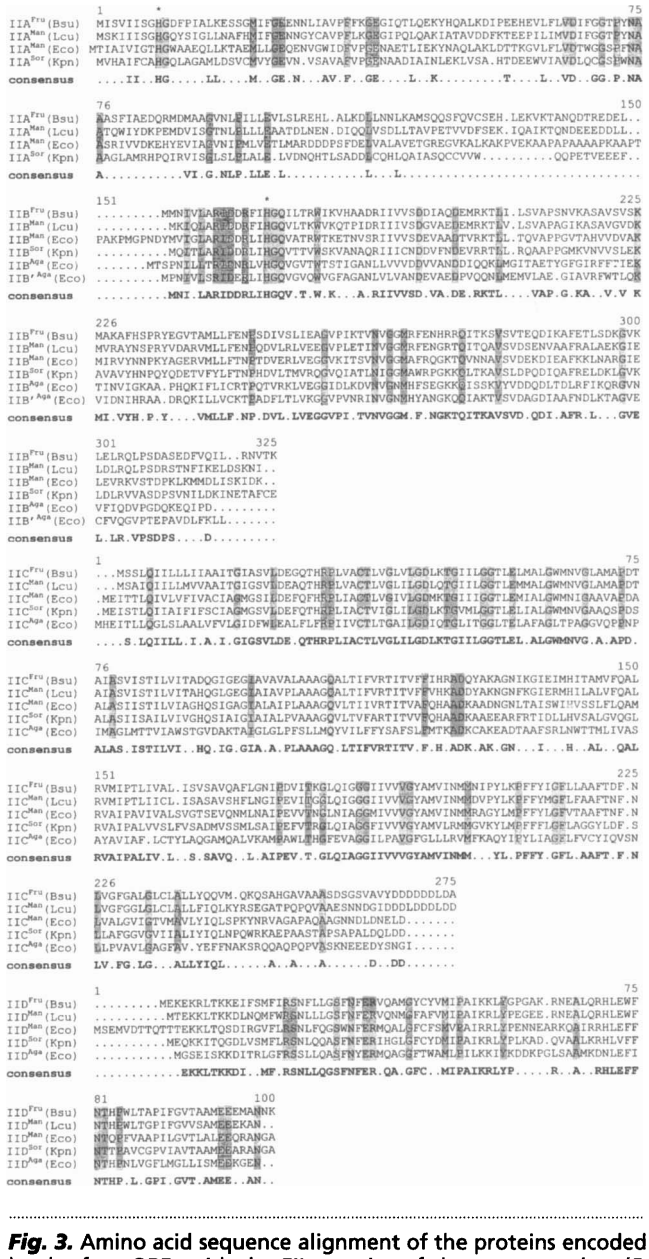


Fig. 3. Amino acid sequence alignment of the proteins encoded by the four ORFs with the EII proteins of the mannose class (*E. coli* EII^{Man} and EII^{Aga}, *K. pneumoniae* EII^{Sor} and *B. subtilis* EII^{Lev}). Fully conserved residues are shaded. Residues common in at least three of the compared proteins are presented in the consensus sequence. The presumed sites of phosphorylation are indicated with an asterisk. Only the first 90 aa of the EIID were used for the alignment.

conserved regions in the EIIC proteins of all four micro-organisms.

Given all those facts, the coding sequences corresponding to the proteins EIIA, EIIB, EIIC and EIID found in the amplified fragments will hereafter be named *manA*, *manB*, *manC* and *manD*.

The existence of a chromosomal copy of the cloned fragments was demonstrated by Southern blotting in which the genomic DNA of *L. curvatus* was digested with *Eco*RI (Fig. 4). In the same experiment, the presence of a similar complex in *L. casei* was tested and its genomic DNA was digested with the endonucleases *Eco*RI and *Bam*HI. The probe used was the 600 bp fragment cloned in pCU2. It hybridized with a band of 2·1 kb from *L. curvatus* DNA. No hybridizing bands were found with *L. casei* DNA (Fig. 4).

Table 2. **Similarity between the A, B, C and D domains of the EII members of the mannose class**

Percentage identities are in parentheses. The entire proteins (domains in the case of the EIIAB^{Man} of *E. coli*) were used for the comparison, except for the comparison of the EIID^{Man} of *L. curvatus* with the other EIID, where only the first 90 aa were used. Eco, *E. coli*; Bsu, *B. subtilis*; Kpn, *K. pneumoniae*; Lcu, *L. curvatus*.

		Man (Eco)	Fru (Bsu)	Sor (Kpn)	Aga (Eco)	Aga′ (Eco)
Man (Lcu)	A	44·2 (25·8)	61·6 (42·5)	54·1 (30·4)	–	–
	B	72·0 (54·0)	80·0 (61·2)	67·9 (46·3)	55·8 (27·9)	58·7 (36·8)
	C	81·0 (60·5)	88·8 (77·3)	77·6 (56·3)	56·5 (24·8)	–
	D	82·8 (58·6)	83·9 (74·7)	72·4 (55·2)	65·1 (38·4)	–
Man (Eco)	A		54·6 (31·2)	48·5 (29·8)	–	–
	B		63·2 (47·2)	66·0 (50·6)	53·5 (33·6)	54·8 (38·1)
	C		79·5 (59·3)	80·1 (60·9)	57·6 (25·8)	–
	D		81·8 (60·4)	81·2 (61·7)	64·1 (33·2)	–
Fru (Bsu)	A			54·8 (29·6)	–	–
	B			65·4 (43·8)	57·1 (27·3)	56·6 (32·5)
	C			77·2 (54·4)	56·9 (25·0)	–
	D			78·1 (57·7)	62·8 (35·6)	–
Sor (Kpn)	A				–	–
	B				51·9 (29·9)	55·8 (34·0)
	C				53·8 (25·8)	–
	D				61·3 (36·4)	–
Aga (Eco)	A					–
	B					69·0 (45·2)
	C					–
	D					–

As discussed above, the sugar specificity pattern of *L. curvatus* was very similar to that previously described in *L. casei* (Veyrat *et al.*, 1994). This hybridization experiment indicates that the sequence divergence between the genes encoding EII within this genus could be high. A second possibility would be that the mannose-glucose-specific EII found in *L. casei* could have a different structure, as described for oral streptococci, in which the type and number of glucose transport systems is variable, even among strains of the same species (Pelletier *et al.*, 1995).

Transcription of the genes encoding EII^{Man}

In an attempt to determine whether the genes encoding EII^{Man} were subject to some type of regulation, total RNA extracted from *L. curvatus* grown on glucose, mannose, fructose, ribose, galactose or maltose was blotted and hybridized with an internal DNA fragment of the mannose complex (pCU2 insert) (Fig. 5). Clear hybridization signals found under all the culture conditions used suggested that these genes are not under tight repression or induction.

Complementation of an EII^{Man} mutant of *L. sake*

The amplified fragments from *L. curvatus* DNA were shown to have a similar sequence and structure to other genes encoding EII^{Man}, but the real function had to be proved. A *Bam*HI–*Hin*dIII fragment from pCU3, containing the *manB* gene from *L. curvatus*, was subcloned in pGAL9. Therefore, *manB* could be expressed in other Gram-positive bacteria from SPO2 and AL9 promoters. A mutant of *L. sake* 160*1K which had been shown to lack a functional EII^{Man}, strain RV37 (Lauret *et al.*, 1996), was transformed with pMJ18. Complemented colonies were selected on MCD medium containing mannose as the only carbon source. The restriction map demonstrated the presence of pMJ18 in the transformants.

The fermentation abilities of *L. sake* 160*1K (wild-type), *L. sake* RV37 and *L. sake* RV37(pMJ18) were tested on Fermentation MRS medium containing glucose, mannose, fructose, ribose and *N*-acetylglucosamine. Acid production of *L. sake* RV37(pMJ18) on these sugars was identical to the wild-type *L. sake* 160*1K. PEP consumption of both strains in the presence of mannose was very similar [24·3 nmol min^{-1} (mg dry wt)$^{-1}$ in the wild-type versus 21·8 nmol min^{-1} (mg dry wt)$^{-1}$ in RV37(pMJ18)], while strain RV37 showed lower activity [2·9 nmol min^{-1} (mg dry wt)$^{-1}$]. Therefore, the gene from *L. curvatus* encoding EIIB^{Man} could complement the impaired EII^{Man} of *L. sake* RV37. This experiment showed that (i) *L. sake* RV37 mutation must be affecting element EIIB^{Man} and that (ii) the sequences described in this work corresponded to the proteins of a true EII^{Man} system belonging to the mannose class.

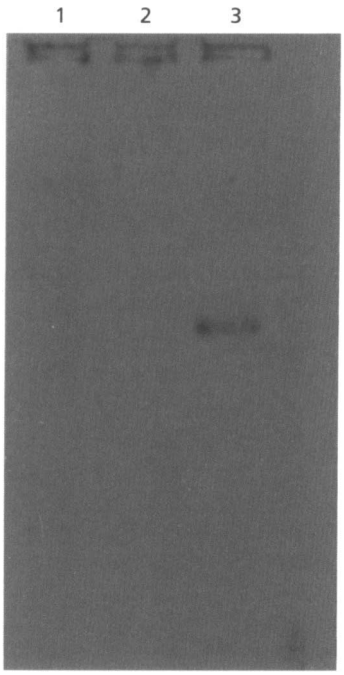


Fig. 4. Southern blot hybridization of total DNA of *L. casei* and *L. curvatus* with the 600 bp fragment as probe. Genomic DNA of *L. casei* was totally digested with two different endonucleases: *Bam*HI (lane 1) and *Eco*RI (lane 2). Total DNA of *L. curvatus* was digested with *Eco*RI (lane 3).

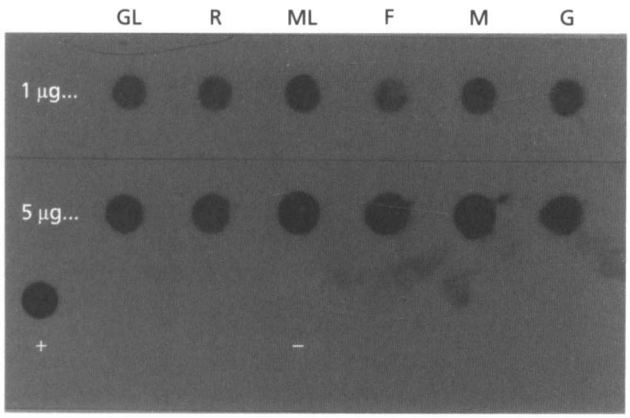


Fig. 5. RNA dot-blot analysis. Total cellular RNA was isolated from *L. curvatus* grown on Fermentation MRS in the presence of 0·5% glucose (G), mannose (M), fructose (F), maltose (ML), ribose (R) and galactose (GL). Each line contained 1 or 5 µg RNA, respectively. The negative control (−) was total RNA from *L. casei*, and the positive control (+) was the fragment (pCU2 insert) used as probe.

Transformation of a *levE* mutant of *B. subtilis*

The levenase operon of *B. subtilis* consists of five inducible genes (*levDEFG* and *sacC*), four of which encode the fructose-EII complex. The fifth gene, *sacC*, encodes a levanase able to hydrolyse sucrose and fructose polymers (Martin *et al.*, 1987, 1989). A model for the induction of the operon has been proposed (Martin *et al.*, 1990). The product of the *levR* gene would undergo a phosphorylation by EII^{Lev}, being inactivated in this form when fructose in the medium is limited. In fact, proteins LevD and LevE ($EIIA^{Lev}$ and $EIIB^{Lev}$) would be involved in this regulation, since the expression of the levanase operon becomes constitutive in mutants deficient in these enzymes. *B. subtilis* QB5072 carries a point mutation in the *levE* gene (G to A) producing the codon change Trp-22 to stop-22. The impaired EII^{Lev} is unable to transport fructose, but the expression of the operon becomes constitutive. Also in this strain, the levanase gene (*sacC*) has been fused to the *lacZ* gene of *E. coli* to facilitate the study of the regulation of this operon.

In order to determine if $EIIB^{Man}$ from *L. curvatus* could have a similar regulatory function to the LevE protein of *B. subtilis*, mutant *B. subtilis* QB5072 was transformed with pMJ18 and pGAL9. When β-galactosidase activity of these transformants was assayed on LB plates containing erythromycin and X-Gal, strains carrying pGAL9 gave dark-blue colonies due to the constitutive expression of the fused operon, while colonies bearing pMJ18 remained white. In order to quantify this activity, both strains were grown on basal medium with glycerol or fructose as carbon sources. β-Galactosidase expression became inducible by fructose in the strain transformed with pMJ18, the activity increasing from 34·8 (with glycerol) to 118·8 (with fructose) U min^{-1} (mg protein)$^{-1}$. The values for QB5072(pGAL9) were 264·3 and 262·8 U min^{-1} (mg protein)$^{-1}$, respectively.

This experiment shows that the constitutive expression of the levanase operon was abolished when the *B. subtilis levE* mutant was transformed with the *manB* gene from *L. curvatus*. The complementation of this regulatory effect suggests that $EIIB^{Man}$ of *L. curvatus* could phosphorylate LevR in *B. subtilis*, as suggested by Stülke *et al.* (1995). Whether $EIIB^{Man}$ is involved in a similar process in *L. curvatus* has yet to be proved. Genetic inactivation techniques which would be required to establish the real role of EII^{Man} in this micro-organism are currently being developed.

Concluding remarks

We have physiologically characterized the glucose transport system and sequenced a DNA fragment of *L. curvatus*, showing that: (i) its sugar specificity pattern corresponds to an EII^{Man}-type of transporter; (ii) fructose is not transported by this system; (iii) ribose is not transported by PTS in this micro-organism; (iv) the amino acid sequence deduced from the fragment amplified by PCR has homology to the other members of the mannose-EII family; (v) sugar phosphorylation as well as dot-blot assays showed that EII^{Man} is constitutive in *L. curvatus*; (vi) $EIIB^{Man}$ from *L. curvatus* can complement a mutant of *L. sake* impaired in EII^{Man}, therefore the cloned operon could encode the PTS transporter of mannose and glucose; (vii) $EIIB^{Man}$ from *L. curvatus* can complement the regulatory functions of LevE from *B. subtilis*, therefore it could play a similar role in *L. curvatus*.

The PTS is known to play a central role, in both the uptake of a number of sugars and regulatory processes, specially by the common elements, HPr and EI. However, the glucose-specific EII elements are also known to mediate in a number of regulatory events (Abe & Uchida, 1989; Veyrat *et al.*, 1994; Martin *et al.*, 1990). Additionally, it has been shown in this work that in *L. curvatus* EII^{Man} could possibly interact with other proteins, such as *B. subtilis* LevE.

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